

H3 (Cohomological Alignment): Arithmetic = Heisenberg

Heisenberg / WH side

Generators $D_{jk}, (j, k) \in (\mathbb{Z}/d\mathbb{Z})^2$

Commutation:

$$[D_{jk}, D_{j'k'}] = \omega^{jk' - j'k} \quad (\omega = e^{2\pi i/d})$$

Central extension:

$$1 \rightarrow \mu_d \rightarrow WH_d \rightarrow (\mathbb{Z}/d\mathbb{Z})^2 \rightarrow 1$$

2-cocycle:

$$c((j, k), (j', k')) = \omega^{jk' - j'k}$$

Class:

$$[c] \in H^2((\mathbb{Z}/d\mathbb{Z})^2, \mu_d)$$

Group + Symplectic Structure

$$G = (\mathbb{Z}/d\mathbb{Z})^2$$

Alternating form:

$$\langle (j, k), (j', k') \rangle = jk' - j'k$$

Bicharacter:

$$e(x, y) = \omega^{jk' - j'k}$$

Properties:

- Alternating
- Bilinear
- Nondegenerate

Cohomological Alignment (H3)

The arithmetic cocycle induced by χ satisfies:

$$[c_\chi] = [c] \in H^2(G, \mu_d)$$

Equivalently, they differ by a coboundary:

$$\exists b : G \rightarrow \mu_d \text{ such that } c_\chi(x, y) = b(x)b(y)b(x+y)^{-1}c(x, y)$$

Failure Modes

Not alternating: introduces symmetric component $\Rightarrow$ wrong cohomology class.

Degenerate: factors through quotient $\Rightarrow$ cannot represent Heisenberg class.